Dodona — cluster sanity

one pass, no load 2026-10-10 16:17 UTC

1. Endpoints

PASS gateway /metrics

PASS gateway /v1/models

PASS prefill /metrics

PASS decode /metrics

2. Model and pools

PASS gateway advertises: Qwen/Qwen2.5-7B-Instruct-AWQ

pool	running	waiting	prompt tok	gen tok
prefill	0	0	18,629	44
decode	0	0	18,629	521

prompt tok 0 on a pool means nothing was ever routed to it.

PASS prefill generated 44 tokens (expected near 0, max_tokens=1)

3. Capacity

quantity	value	where it comes from
KV per decode pod	4.30 GiB	pod log: Available KV cache memory
full-length seqs / pod	4.9	pod log: Maximum concurrency
decode replicas	1	kubectl get deploy vllm-decode
decode pool concurrency	4.9	seqs/pod x replicas

A pool this small is the point: a policy needs something to be scarce.

PASS pool of 4.9 is small enough for KV to bind

4. Does a request traverse the gateway

PASS orch_requests_total moved 46 -> 47

PASS completion returned in 0.305s

field	value
content	'ok'
via	local
latency	0.305s (tunnel included)

via=local means our cluster served it; via=overflow means it was bursted out.

5. The KV hop

PASS the router recorded a hop

field	value
src	prefill-0
dst	decode-0
req_id	http
tokens	16
backend	mooncake
prefix	802848469d1623df

FAIL connector transport is known: unknown

5b. Does decode re-prefill?

pool	prompt tok	prefix hits	queries	hit rate
prefill	18,665	10,272	18,665	55.0%
decode	18,665	10,272	18,665	55.0%

Decode loading blocks rather than recomputing shows as a high hit rate.

FAIL decode processes fewer prompt tokens than prefill

Both pools report ~18,665 prompt tokens. queue.py sends the same prompt to both legs, so this is expected — and it means decode is NOT skipping prompt work. Confirm with TTFT on a long prompt.

6. Was anything scarce

pool	kv_cache_usage_perc	vs 0.90 threshold
prefill	0.0000	BELOW
decode	0.0000	BELOW

Idle cluster: near zero is expected. make load is what moves this.

An idle reading is not a verdict on the policy — see make load.

7. Where the work lands

7b. The slice budget

component	GiB
slice x util	10.80
weights	5.20
activations + CUDA graphs	1.30
KV cache	4.30

Overhead is budget minus weights minus KV. Measured, not predicted.

8. Latency, short prompt

metric	seconds
n	10
min	0.285
p50	0.295
p95	0.306
max	0.306

Through the tunnel. Shape only — use make load to quote numbers.

9. Latency vs prompt length

prompt tokens	p50 seconds

Warmup discarded, so these are cache-warm numbers.

FAIL all 5 prompt lengths completed

256 tokens DROPPED — HTTP 429: {"error": {"type": "rate_limit_error", "message": "tenant tokens/min cap"}, "via": "local
1,024 tokens DROPPED — HTTP 429: {"error": {"type": "rate_limit_error", "message": "tenant tokens/min cap"}, "via": "loc
4,096 tokens DROPPED — HTTP 400: {"error": {"type": "prompt_too_long", "message": "prompt > 8000 chars"}}
8,192 tokens DROPPED — HTTP 400: {"error": {"type": "prompt_too_long", "message": "prompt > 8000 chars"}}
12,288 tokens DROPPED — HTTP 400: {"error": {"type": "prompt_too_long", "message": "prompt > 8000 chars"}}

10. Grafana

vLLM — KV, running/waiting, TTFT, prefix cache

`http://127.0.0.1:31495/d/dodona-vllm?from=now-30m&to=now`

Gateway — admit, shed, place, latency by stage

`http://127.0.0.1:31495/d/dodona-gateway?from=now-30m&to=now`

Router — pick, sticky, KV hop

`http://127.0.0.1:31495/d/dodona-router?from=now-30m&to=now`

KEDA — scaler value vs replicas

`http://127.0.0.1:31495/d/dodona-keda?from=now-30m&to=now`

HAMi — device memory vs the slice cap

`http://127.0.0.1:31495/d/dodona-hami?from=now-30m&to=now`

explore: KV usage per pod

`http://127.0.0.1:31495/explore?left=%7B%22datasource%22%3A%20%22Prometheus%22%2C%20%22queries%22%3A%20%5B%7B%22`

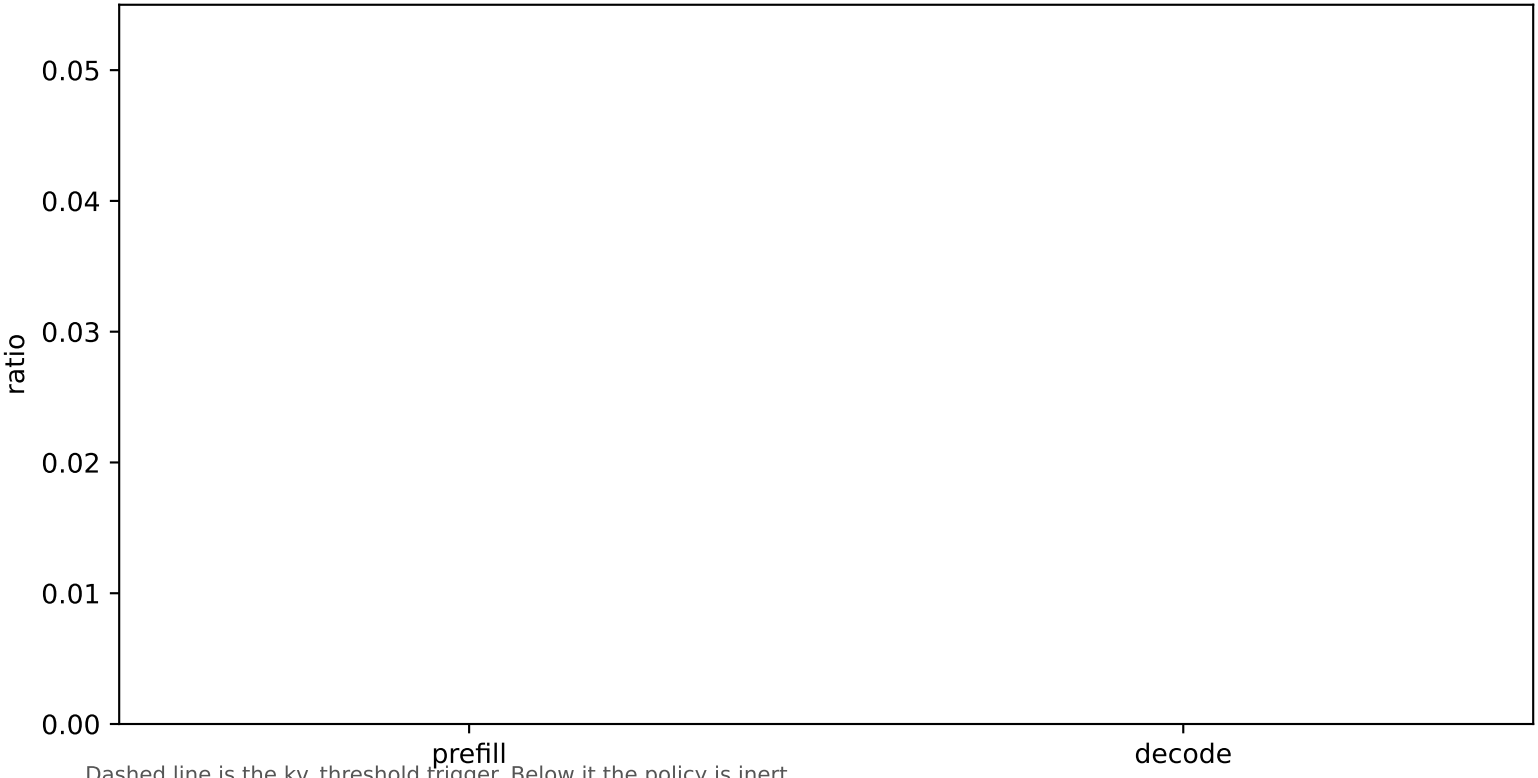
explore: queue depth per pod

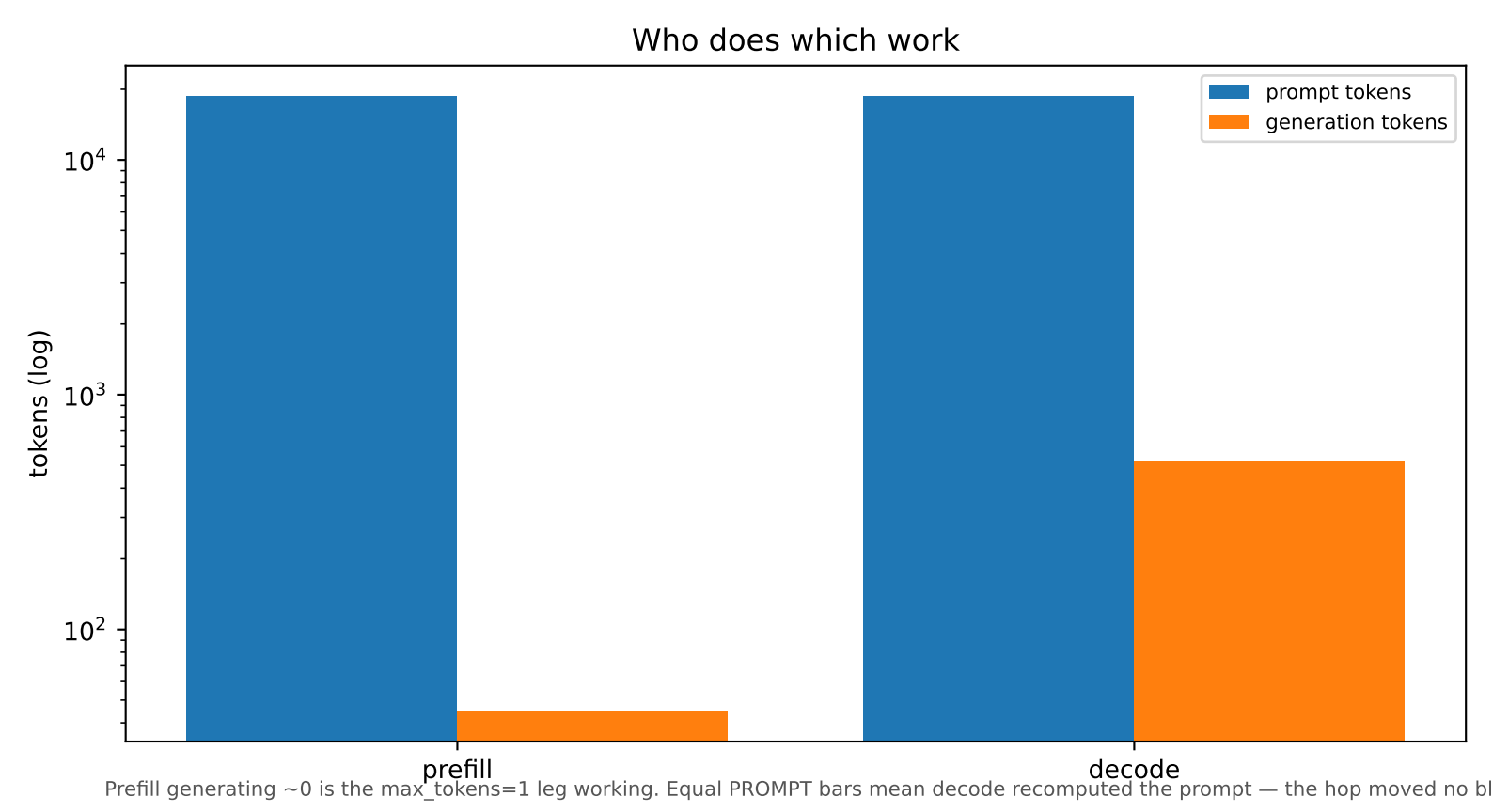
`http://127.0.0.1:31495/explore?left=%7B%22datasource%22%3A%20%22Prometheus%22%2C%20%22queries%22%3A%20%5B%7B%22`

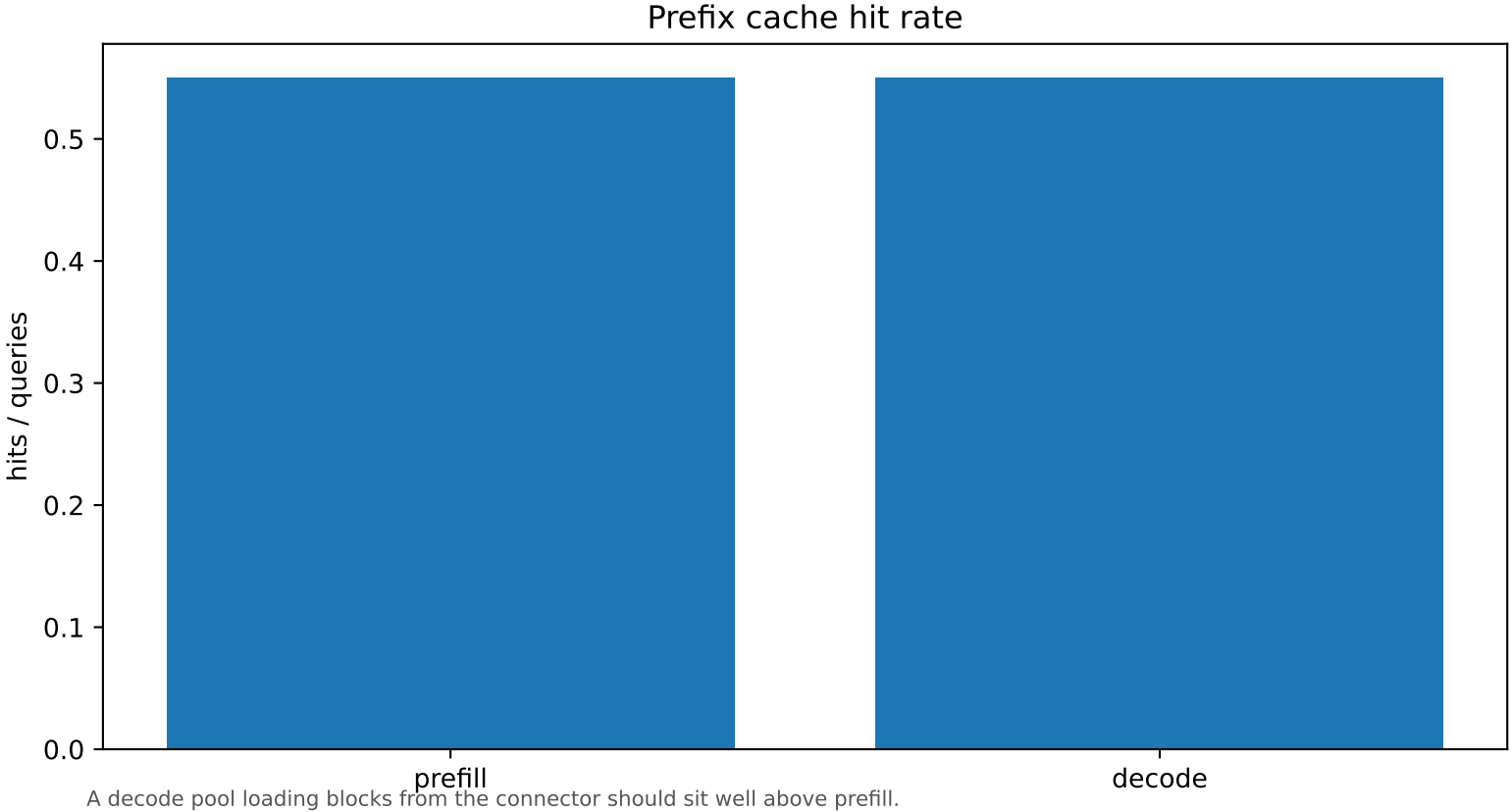
explore: gateway shed by reason

`http://127.0.0.1:31495/explore?left=%7B%22datasource%22%3A%20%22Prometheus%22%2C%20%22queries%22%3A%20%5B%7B%22`

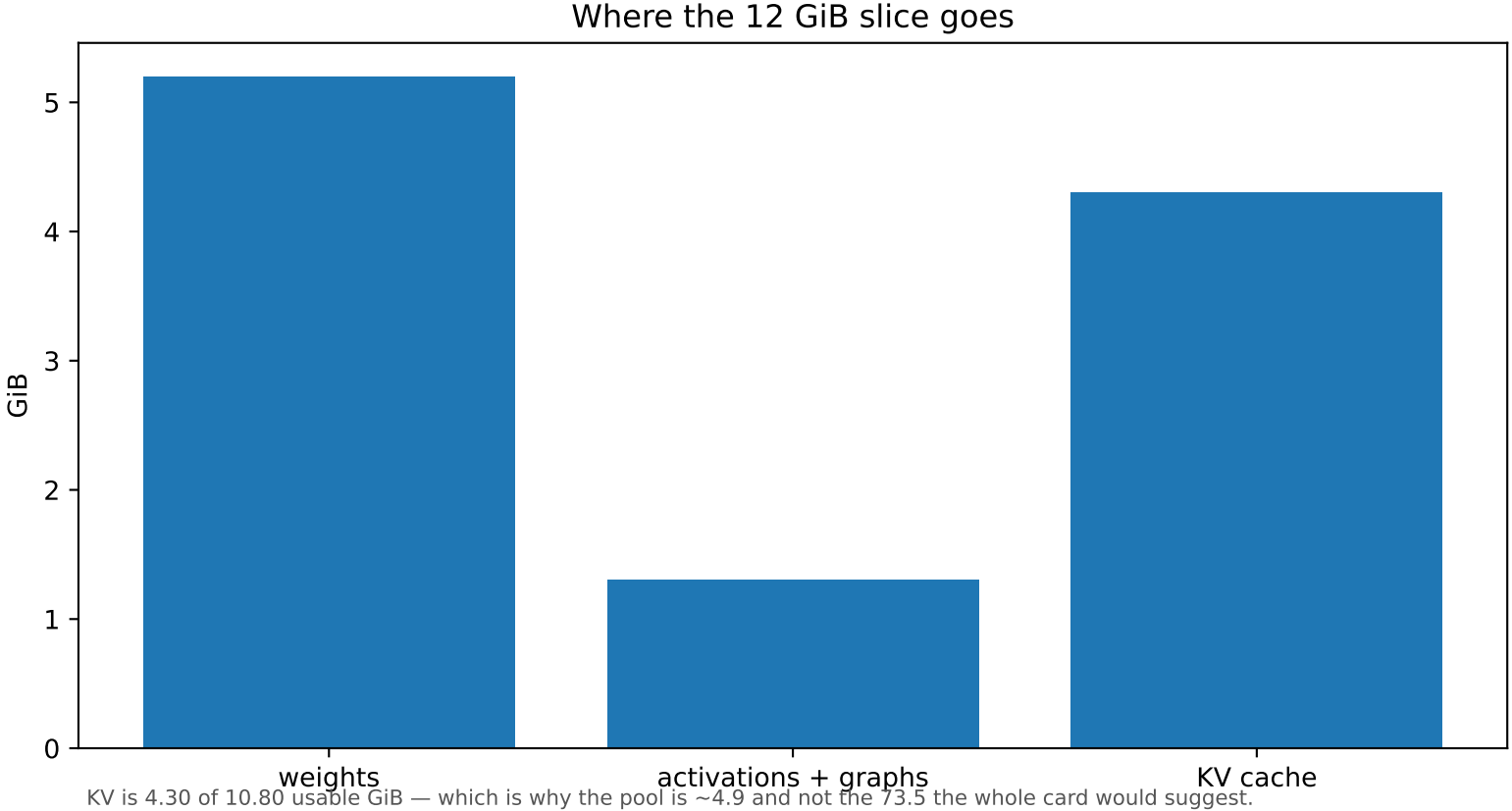
Grafana password is printed by make observability.

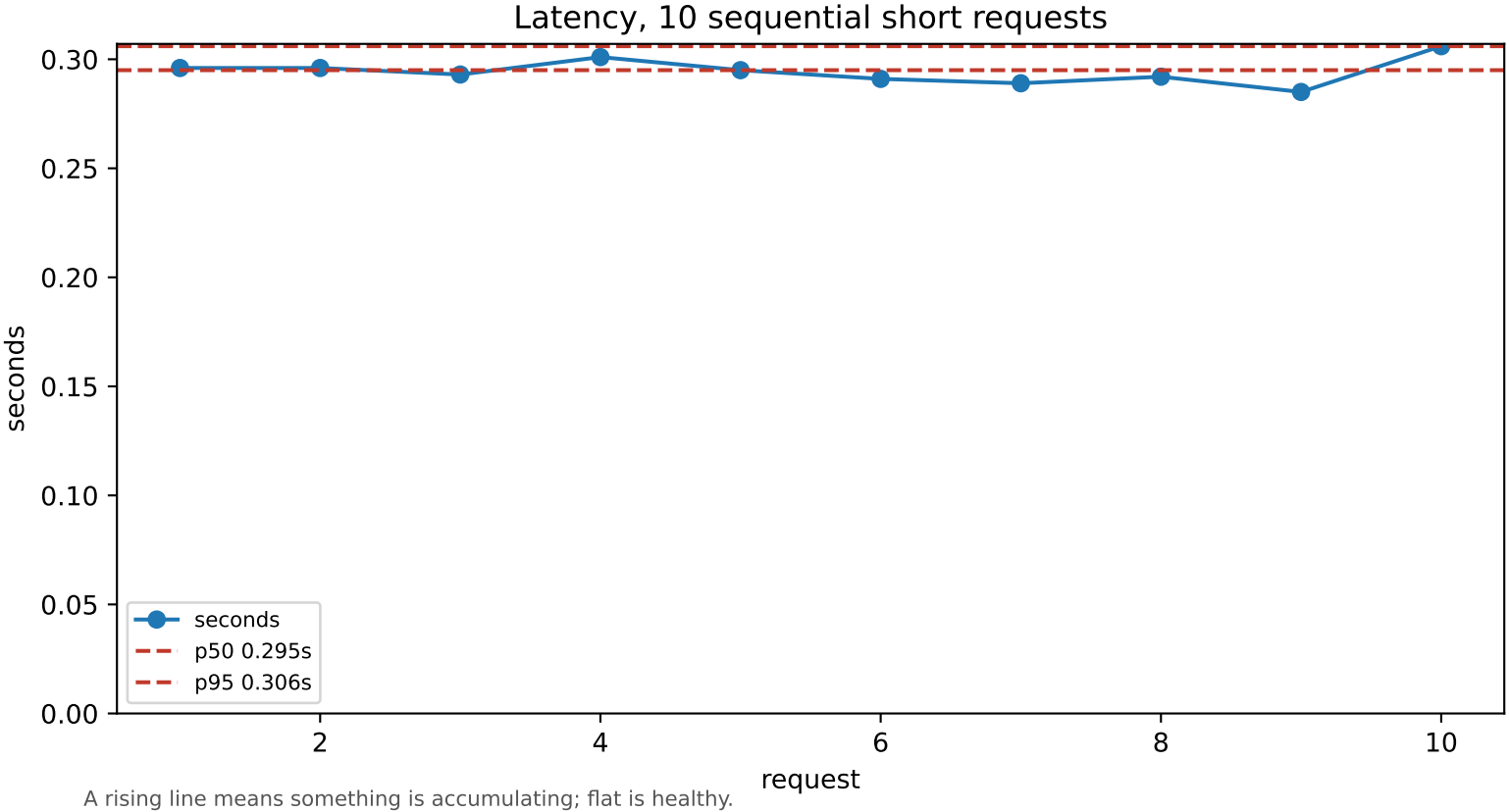

KV usage vs the 0.90 admission threshold
ratio
0.05
0.04
0.03
0.02
0.01
0.00
prefill
decode
Dashed line is the kv_threshold trigger. Below it the policy is inert.


Who does which work
prompt tokens
generation tokens
tokens (log)
10^4
10^3
10^2
prefill
decode
Prefill generating ~0 is the max_tokens=1 leg working. Equal PROMPT bars mean decode recomputed the prompt — the hop moved no bl



A decode pool loading blocks from the connector should sit well above prefill.


Where the 12 GiB slice goes
GiB
0
1
2
3
4
5
weights
activations + graphs
KV cache
KV is 4.30 of 10.80 usable GiB — which is why the pool is ~4.9 and not the 73.5 the whole card would suggest.



A rising line means something is accumulating; flat is healthy.